PhET- Density Activity

Custom Section
Name_____

Material	Mass (kg)	Volume (L)	Density (kg/L)	Does it Float?
Styrofoam				
Wood				
Ice				
Brick				
Aluminum				

1. In the custom setting, choose the 'My Object' option in the material drop down box. Set the mass of your object to 4 kg. Adjust the volume to find the minimum volume needed to make the object float.

Volume_____ Density_____

Same Mass Section

Material	Mass (kg)	Volume (L)	Density (kg/L)	Does it Float?
Blue				
Yellow				
Green				
Red				

Same Volume Section

Material	Mass (kg)	Volume (L)	Density (kg/L)	Does it Float?
Blue				
Yellow				
Green				

Red				

3. Looking at the data, what must be true about the density of an object in order for it to float?

Same Density Section:

4. Calculate the density of the blue object in this section.

Mass _____ Volume_____
Density_____

5. Explain why both the yellow and red objects float when they have different sizes.

Mystery Section:

6. Before you start, pick an object that you think will float. _____
Pick an object that you think will sink. _____

Material	Mass (kg)	Volume (L)	Density (kg/L)	Does it Float?
A				
B				
C				
D				
E				

7. In the Custom section describe the difference between how Styrofoam and ice floated. Also explain why you think this is the case?

8. In the Same Mass Section discuss what was interesting about the blue object's behavior in the water.

9. In the Mystery Section, click on the “Show Table” button. What is the most dense
object on the list? Write its density as well.